ADI Internship

Mixer Assignment

Prepared by

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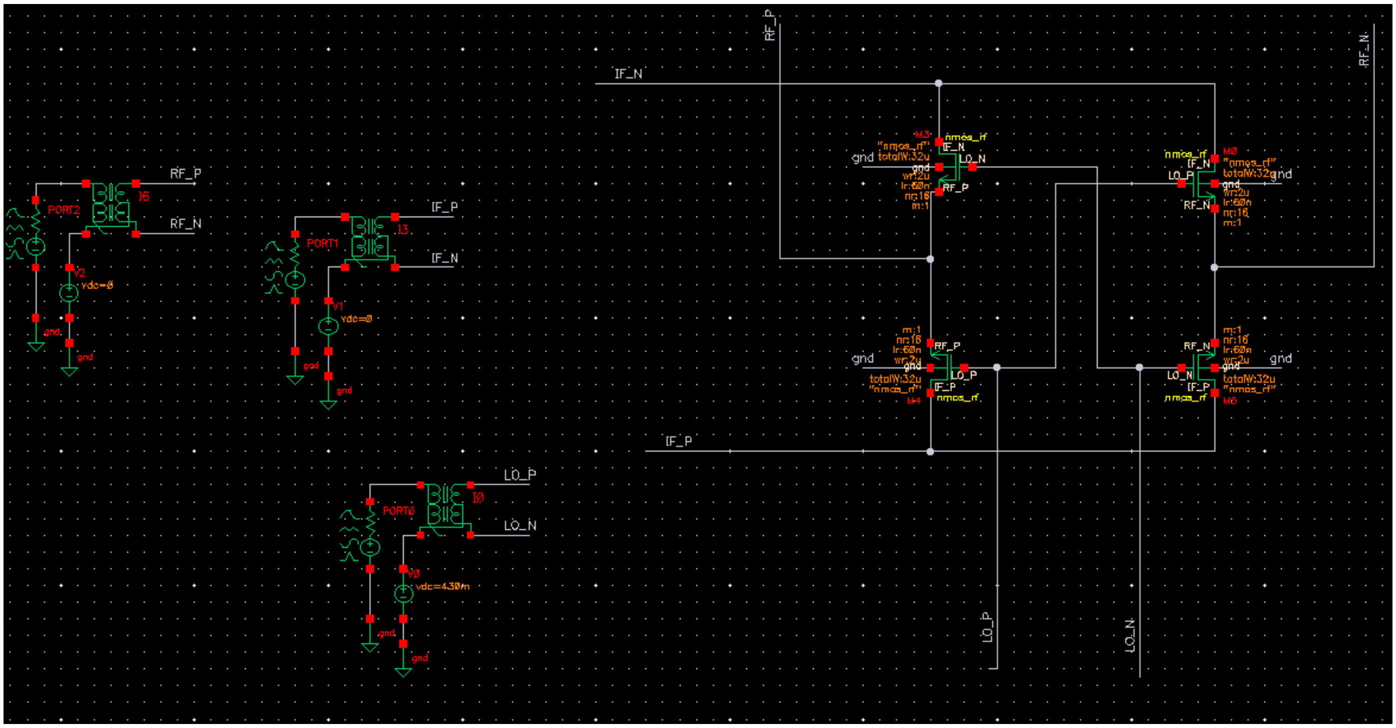
Mixer Assignment

1. Schematic

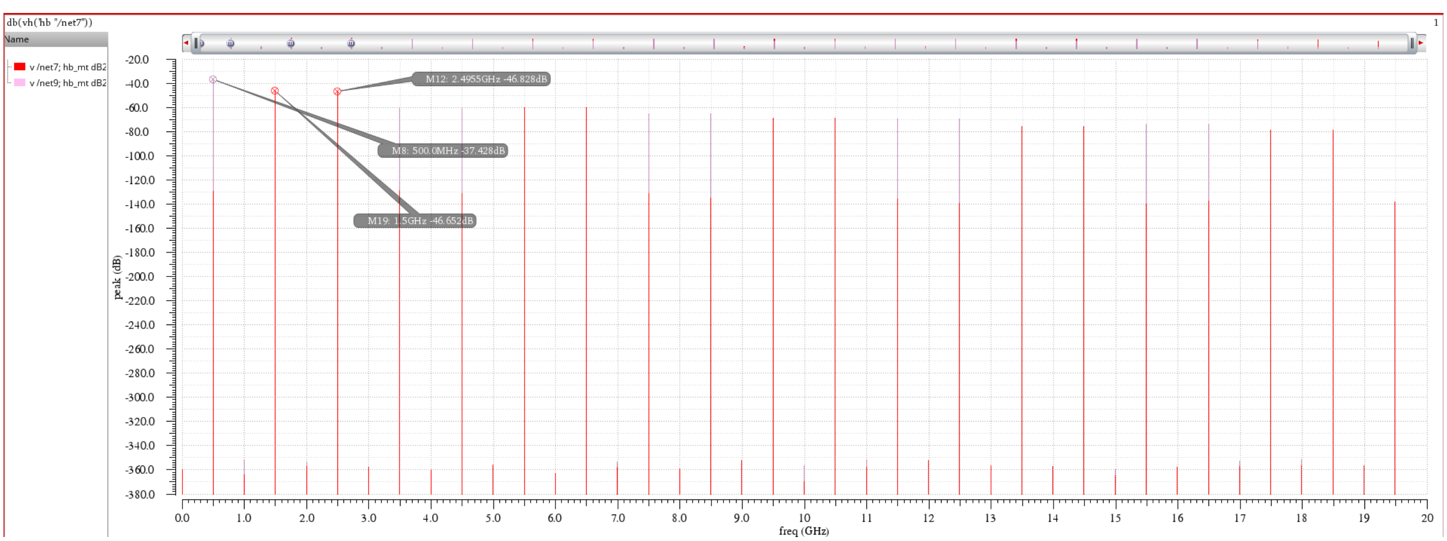


- Double Balanced feedthrough mixer
- 2 input ports for IF and LO and one output port RF (up converter)
- Common mode value for the gate is equal to 430m (V_{th}) to improve switching (by decreasing R_{on})
- Common mode value for the source and drain is zero ($V_{DS} = 0$) to work in linear region (as a switch)

2. Harmonic Balance Analysis

<u>Spectrum for showing input and output harmonics</u>

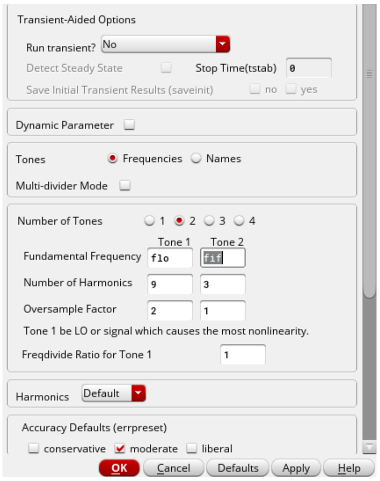
- The input IF and output Both up converted and down converted are marked



- The up converted component is extracted using harmonic indexing.

	Expression	Value
1	db(vh('hb "/net...	-46.65

- <u>The harmonic balance analysis setup</u>



<u>Calculation Expressions</u>

mixer:up_conv:1	V_CG	expr	(V_RF_dBV + 30 + 10)
mixer:up_conv:1	V_RF	expr	mag(vh('hb "/net7" '(1 1)))
mixer:up_conv:1	V_RF_dBV	expr	(20 * log10(V_RF))

<u>Parameters used</u>

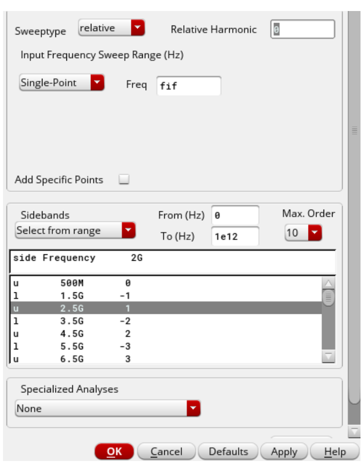
flo	2G
fif	0.5G
pif	-30
plo	10

- **The Results (The Conversion Gain)**

mixer:up_conv:1	V_RF	4.648m
mixer:up_conv:1	V_RF_dBV	-46.65
mixer:up_conv:1	V_CG	-6.654

3. Harmonic Balance AC Analysis

- <u>Harmonic Balance AC Analysis setup</u>



<u>Calculation Expressions</u>

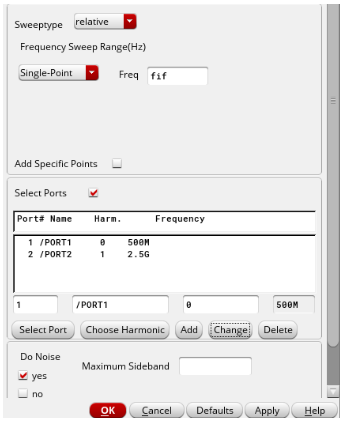
mixer:up_conv:1	V_CG	expr	(V_RF_dBV + 30 + 10)
mixer:up_conv:1	V_RF	expr	mag(vh('hbac "/net7" '(1)))
mixer:up_conv:1	V_RF_dBV	expr	(20 * log10(V_RF))

- **The Results (The Conversion Gain)**

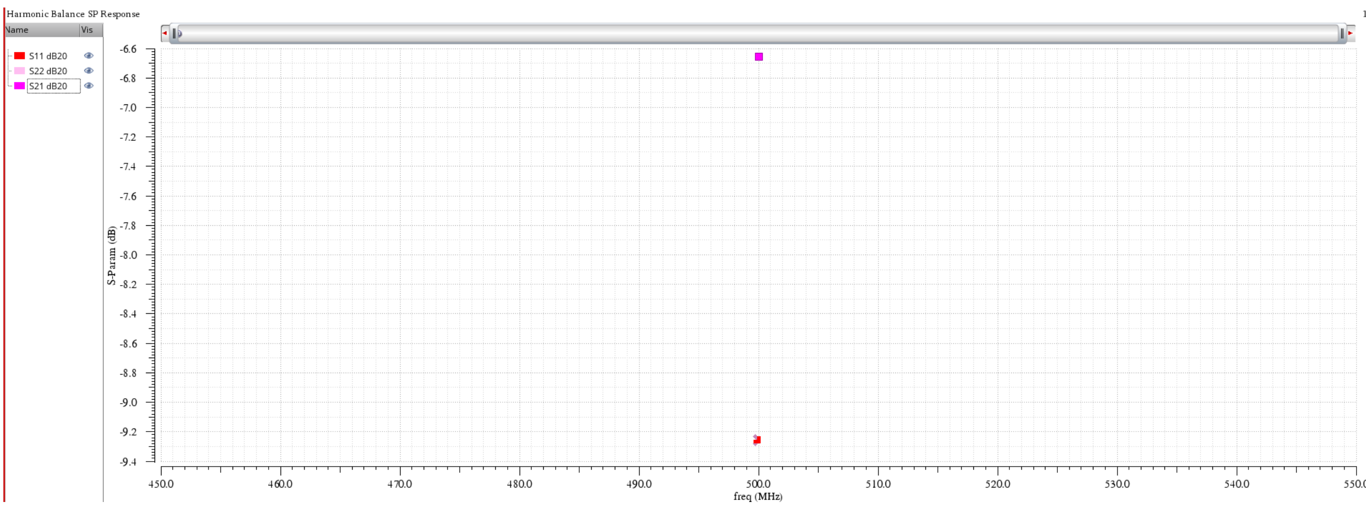
mixer:up_conv:1	V_RF	4.651m
mixer:up_conv:1	V_RF_dBV	-46.65
mixer:up_conv:1	V_CG	-6.649

4. Harmonic Balance S-parameters Analysis

- <u>Third Harmonic Balance S-parameters Analysis setup</u>



- Showing S_{11} , S_{22} and S_{21} on the spectrum



- **The results (S_{11} , S_{22} and S_{21} $(conversion\ gain)$)**

	freq (Hz)	S11 dB20 (dB)	S22 dB20 (dB)	S21 dB20 (dB)
1	500.0E6	-9.263	-9.254	-6.649

5. Comparison

Analysis	Conversion Gain (dB)
HB	-6.654
HBAC	-6.649
HBSP	-6.649

✓ Almost the conversion gain obtained from all tests is the same